Pipeline Analysis Results

Atlas: AAL (90 nodes) **Subjects:** 50 **Sessions:** 3 **Groups:** 5

Groups detected: control, intervention_alone_short, intervention_alone_long, intervention_group_short, intervention_group_long

Script	Status	Key Output
global_basic_metrics	✓ Complete	global_metrics.parquet
nodal_hub_identification	✓ Complete	node_level_metrics.parquet
nodal_temporal_trajectories	✓ Complete	node_trajectories.parquet
nodal_multivariate_analysis	✓ Complete	five_group_anova.parquet
stat_svm_baseline_analysis	■ Fix needed	AttributeError: run_spec
stat_svm_time_5groups	✓ Complete	svm_rf_time_summary.json
stat_svm_stratified_5groups	✓ Complete	stratified_svm_summary.json
stat_random_forest_comparison	✓ Complete	rf_svm_comparison_summary.json

1. Global Network Metrics

Atlas: AAL | 90 nodes | Format: .npy | Sessions: [1, 2, 3]

1.1 Group Comparison

Group	N	Global Efficiency	Clustering Coef	Path Length
control	30	0.155 ± 0.001	0.140 ± 0.001	6.467 ± 0.027
intervention_alone_short	30	0.158 ± 0.002	0.144 ± 0.003	6.340 ± 0.083
intervention_alone_long	30	0.160 ± 0.003	0.147 ± 0.004	6.268 ± 0.107
intervention_group_short	30	0.159 ± 0.002	0.145 ± 0.003	6.309 ± 0.091
intervention_group_long	30	0.161 ± 0.004	0.148 ± 0.005	6.228 ± 0.135

Observation: Intervention groups show progressive increase in global efficiency and clustering coefficient across sessions, with longer interventions showing stronger effects (long > short).

2. Nodal Hub Identification

13,500 node records | 150 subject-sessions | 90 nodes per session

2.1 Hub Type Distribution

Hub classification per node using Guimera & Amaral (2005) criteria:

Hub Type	Threshold	Description
Provincial Hub	z-score > 2.5, PC < 0.30	High within-module z-score, low participation
Connector Hub	PC > 0.30, z-score > 1.0	High participation, moderate z-score
Kinless Node	PC in [0.05, 0.30]	Intermediate participation
Peripheral	PC < 0.05, z-score < 1.0	Low connectivity, local role

Plots generated: hub_counts_by_group.png, participation_by_group.png, hub_type_distribution.png

3. Nodal Temporal Trajectories

3,150 trajectories | 630 node × metric effects | 90 nodes × 7 metrics

3.1 Top Responding Nodes

Rank	Node	Metric	Cohen's d	Interpretation
1	39	clustering	17.36	Very strong intervention effect
2	39	local_efficiency	17.11	Very strong intervention effect
3	21	clustering	14.45	Very strong intervention effect
4	21	local_efficiency	14.34	Very strong intervention effect
5	31	strength	13.77	Very strong intervention effect

3.2 Effect Size Summary

Threshold	Count	Percentage
Strong effect (|d| > 0.5)	438	69.5%
Moderate effect (|d| > 0.2)	501	79.5%

Group detection: Auto-detected — control vs [alone_short, alone_long, group_short, group_long]

4. Nodal Multivariate Analysis

630 node-metric combinations analyzed across 6 statistical analyses

Analysis	Total	Significant ($p<0.05$)	%	Large Effect	%
5-Group ANOVA	630	306	48.6%	222 ($\eta^2>0.14$)	35.2%
Social Effects	630	70	11.1%	0 ($\|d\|>0.8$)	0.0%
Duration Effects	630	202	32.1%	5 ($\|d\|>0.8$)	0.8%
Intervention/Control	630	296	47.0%	218 ($\|d\|>0.8$)	34.6%
Gender Effects	630	0	0.0%	—	—
Age Correlations	—	—	—	—	—

Note: Gender Effects = 0 because sex column not propagated to node_level_metrics.parquet. Age Correlations skipped — age column not found in parquet (fix: re-run nodal_hub_identification with updated mock data including age).

5. SVM + Random Forest Classification

50 subjects | 9 features (7 metric slopes + sex + n_sessions)

5.1 Time-Effect Analysis (stat_svm_time_5groups)

Model	CV Accuracy	Train Accuracy	Best
Random Forest	0.580 ± 0.075	0.920	
SVM (RBF)	0.560 ± 0.162	0.780	✓

5.2 Top ANOVA Features

Feature	F-Score	p-value
strength_slope	75.07	2.5e-19
clustering_slope	74.06	3.3e-19
local_efficiency_slope	73.50	3.8e-19
betweenness_slope	15.15	6.5e-08
participation_coef_slope	1.71	0.164

5.3 Stratified Analysis (stat_svm_stratified_5groups)

Stratum	N	CV Accuracy	Train Accuracy
Female	19	0.700 ± 0.187	0.842
Male	31	0.610 ± 0.091	0.806

6. RF vs SVM Comparison (stat_random_forest_comparison)

Variant	RF CV	SVM CV	Best Model
Social	0.700 ± 0.009	0.800 ± 0.026	SVM
Duration	0.800 ± 0.073	0.803 ± 0.119	SVM
Intervention	1.000 ± 0.000	1.000 ± 0.000	RF (tie)

Intervention vs Control = 100% — Mock data is perfectly separable by design. Expected to be lower with real data.

Top ANOVA Features across Variants

Feature	Social F	Duration F	Intervention F
clustering_slope	75.66	93.78	115.26
local_efficiency_slope	74.09	93.04	112.98
strength_slope	72.16	99.49	114.01
betweenness_slope	31.44	29.37	59.74

7. Issues & Next Steps

7.1 Known Issues

Issue	Severity	Fix
stat_svm_baseline_analysis: AttributeError run_spec	Medium	Add run_spec=None in main()
Age column not in node_level_metrics.parquet	Low	Re-run nodal_hub with updated mock data
Gender Effects = 0 in multivariate	Low	Sex not propagated through aggregation
codecovcli blocked by Windows policy	Low	Use python -m codecov_cli instead

7.2 Next Steps

1. Fix stat_svm_baseline_analysis (run_spec AttributeError)
2. Re-generate mock data with age column → re-run nodal pipeline
3. Run codecov upload with python -m codecov_cli
4. Test with real data (Test_matrizen)
5. Add 5_responder_analysis scripts
6. Add 6_visualization scripts
7. GitHub Actions für automatischen Codecov Upload